THE SQUARE GRID, INVERSES AND PRIMES THAT SPLIT

Gaussian integers: a ring of squares

$13 = 3^2 + 2^2$ becomes a factorisation once the square grid becomes a number system.

1 / WHICH OPERATIONS STAY INSIDE?

A commutative ring; division can leave it

$$\mathbb{Z}[i] = \{a + b\mathrm{i} : a, b \in \mathbb{Z}\}, \qquad \mathrm{i}^2 = -1.$$

Addition and subtraction keep integer coordinates. So does multiplication:

$$(a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}.$$

Addition and multiplication are associative and commutative; multiplication distributes over addition. With $0, 1$ and additive inverses, this is a **commutative ring**.

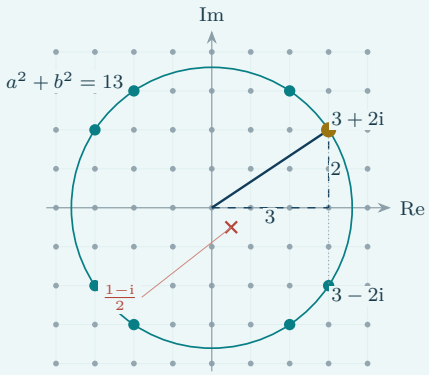
A **field** also gives every non-zero element an inverse within the set. Here, for example,

$$\frac{1}{1+\mathrm{i}} = \frac{1-\mathrm{i}}{2} \notin \mathbb{Z}[i].$$

Thus $\mathbb{Z}[i]$ is **not a field**. Allowing rational coordinates gives $\mathbb{Q}(i) = \{a+b\mathrm{i} : a, b \in \mathbb{Q}\}$, which is a field.

2 / NUMBERS BECOME LATTICE POINTS

The point $3 + 2\mathrm{i}$ is $(3, 2)$



Dots: Gaussian integers.
The cross marks an inverse off the grid.
The circle passes through eight lattice points.

3 / A WHOLE-NUMBER MEASURE OF SIZE

The norm multiplies

For $z = a + b\mathrm{i}$, its conjugate is $\bar{z} = a - b\mathrm{i}$:

$$\mathrm{N}(z) = z\bar{z} = a^2 + b^2 = |z|^2, \qquad \mathrm{N}(zw) = \mathrm{N}(z)\,\mathrm{N}(w).$$

The norm is the **squared distance** from the origin. If $zw = 1$, the two non-negative integer norms must multiply to 1. Hence the only **units**, the elements invertible inside $\mathbb{Z}[i]$, are

$$\mathrm{N}(z) = 1 \iff z \in \{1, -1, \mathrm{i}, -\mathrm{i}\}.$$

Multiplying by a unit rotates a lattice point through a multiple of $90°$.

4 / FERMAT PRIMES

An ordinary prime can acquire factors

$$13 = 3^2 + 2^2 = (3 + 2\mathrm{i})(3 - 2\mathrm{i}).$$

Each factor has norm 13, so neither is a unit. Thus 13, prime in $\mathbb{Z}$, is composite in $\mathbb{Z}[i]$.

A **Gaussian prime** here means a non-zero non-unit that cannot be factored into two non-units. Both $3 + 2\mathrm{i}$ and $3 - 2\mathrm{i}$ are Gaussian primes: their prime norm 13 forces one factor in any factorisation to have norm 1.

Notice $\mathrm{N}(3 + 2\mathrm{i}) = 13$, whereas $\mathrm{N}(13) = 169$.

5 / CLASSIFY THE ORDINARY PRIMES

Fermat's two-square theorem and three Gaussian behaviours

For an ordinary positive prime p,

$$p = a^2 + b^2 \text{ for some integers } a, b \iff p = 2 \text{ or } p \equiv 1 \pmod 4.$$

ordinary prime	example or factorisation	behaviour in $\mathbb{Z}[i]$
$p = 2$	$2 = -\mathrm{i}(1 + \mathrm{i})^2$	repeated prime factor: **ramifies**
$p \equiv 1 \pmod 4$	$13 = (3 + 2\mathrm{i})(3 - 2\mathrm{i})$	two conjugate prime factors: **splits**
$p \equiv 3 \pmod 4$	$3, 7, 11, 19, \ldots$	remains Gaussian prime: **inert**

Here $p \equiv r \pmod 4$ means that p leaves remainder r on division by 4. For 2, the factors $1 + \mathrm{i}$ and $1 - \mathrm{i} = -\mathrm{i}(1 + \mathrm{i})$ differ by a unit.

A prime norm guarantees a Gaussian prime. A Gaussian prime can also have composite norm: 3 remains prime although $\mathrm{N}(3) = 9$.

6 / WHY DOES FOUR APPEAR?

An easy exclusion; a deeper converse

A square leaves remainder 0 or 1 modulo 4. A sum of two squares can therefore leave $0, 1$ or 2, never 3. This rules out every prime $p \equiv 3 \pmod 4$ as such a sum.

Proving that every prime $p \equiv 1 \pmod 4$ works needs more. A Gaussian proof uses **Euclidean division**: rounding a complex quotient to the nearest lattice point gives a remainder of smaller norm. Euclid's algorithm and unique factorisation follow, with factor order and units allowed to vary.

7 / BACK TO PYTHAGOREAN TRIPLES

Square a lattice point

$$(3 + 2\mathrm{i})^2 = 5 + 12\mathrm{i}.$$

Taking norms gives

$$5^2 + 12^2 = \mathrm{N}\big((3 + 2\mathrm{i})^2\big) = 13^2.$$

For integers $m > n > 0$,

$$(m + n\mathrm{i})^2 = (m^2 - n^2) + 2mn\mathrm{i},$$

so $\mathrm{N}(z^2) = \mathrm{N}(z)^2$ produces the homogeneous point

$$(m^2 - n^2 : 2mn : m^2 + n^2).$$

Read further: Oxford, *Rings & Arithmetic 9: The Gaussian integers*; *Number Theory: In Context*, §14.1.

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